

Testing Guide

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Run tests from `06Code` :

```
npm install
npm test
```

Run the API validation report:

```
npm run test:api:validation
```

Run the Postman/Newman JWT evidence:

```
npm run postman:evidence:local
```

Current automated coverage:

- Auth/session: registered Google login, unregistered Google rejection, unverified Google email rejection, Google email linking, linked-email mismatch rejection, inactive account rejection, email/password login, seeded password-hash role login, temporary password change enforcement, director-created account login, invalid branch assignment rejection, logout, invalid Google token, expired session, private page redirect/no-store and valid private dashboard access.
- JWT/Postman session: `/auth/login` returns `sessionToken`, Bearer token can call `/auth/me`, logout revokes the backend session, and the same token returns `401`.
- Cache management: public auth config cache headers, no-store session responses, private catalog memory-cache `MISS/HIT`, branch-list invalidation

after writes and scoped branch report cache.

- RBAC: invalid role cannot access consolidated reports.
- Actor flows: Visitor enrollment, Student first-password change and own-scope access, Teacher attendance work, BranchDirector branch scope, GeneralDirector account/branch-access management, and Admin governance.
- Rules: scholarship threshold, promotion candidate rule, teacher payment calculation.
- Academic integration: enrollment request, duplicate attendance rejection, absence review, scholarship evaluation, level promotion evaluation.
- Python analytics: health cache headers, protected no-store responses and analytics service calculations.

Expected result:

```
Test Suites: 6 passed, 6 total
Tests:      38 passed, 38 total
```

Current API validation result:

```
API validation completed: 122/122 passed.
```

Current Python analytics test result:

```
Ran 9 tests in 0.042s
OK
```

Current Postman/Newman evidence:

```
Requests: 67 executed, 0 failed
Assertions: 78 executed, 0 failed
```

Evidence files:

- 03Documentation/cache-management.md
- 03Documentation/api-validation-report.md
- 03Documentation/actor-flows/
- 070ther/api-validation-results.json
- postman/evidence/postman-local-jwt-auth-evidence.md
- postman/evidence/postman-local-jwt-auth-evidence.json

If npm reports `UNABLE_TO_VERIFY_LEAF_SIGNATURE` in a local/corporate Windows environment, configure the local certificate chain or run the install in a trusted shell configuration. Do not disable TLS verification in production.